A. Uncertainty Quantification for Primary Results

In this section we quantify uncertainty over the results presented in Section 4. To do this, we follow Karger et al. (2024) in computing pairwise p-values via bootstrap resampling. In particular, we first compute the mean difference in Brier scores at the individual question level, which will serve as our statistic of interest. This is equivalent to simply computing the difference in overall Brier score that two different methods achieve on a given benchmark. We then approximate the null distribution of this statistic by (1) recentering these question-level differences at zero, (2) drawing 10000 bootstrap samples from these recentered differences, and (3) computing the mean difference in Brier score for each of these samples. This procedure will, for example, have power to detect cases where one forecaster performs slightly better than another *uniformly* over all questions, even if the per-question performance gap is small (e.g., the better forecaster always provides a forecast that is 1% closer to the true outcome). In contrast, computing separate confidence intervals for each Brier score, without conditioning on the questions, will struggle to detect this kind of performance gap. Importantly, this procedure does not account for uncertainty over the LLM outputs themselves. We address this separately Section 7. We present these results in Tables 12, 13, 14 and 15 below.

Forecaster	Brier score	Win Rate vs Best	Pairwise p-value
Market	0.0956	0.4212	0.0687
Public Survey (median per-event)	0.1035	0.2236	<0.001
Karger et al. (2024) state of the art	0.107	-	-
OpenAI o3	0.1096	0.6447	0.0305
AIA Forecaster	0.0753	0.3026	0.4328
Superforecasters (median per-event)	**0.0740**	-	-

Table 12: Uncertainty quantification for results on FB-Market in Table 2. Karger et al. (2024) do not publish individual LLM forecasts, precluding a direct pairwise comparison.

Forecaster	Brier score	Win Rate vs Best	Pairwise p-value
Public Survey (median per-event)	0.1451	0.2149	<0.001
Superforecasters (median per-event)	0.1110	0.3474	0.1522
Karger et al. (2024) state of the art	0.133	-	-
OpenAI o3	0.1221	0.4839	0.0025
AIA Forecaster	**0.1076**	-	-

Table 13: Uncertainty quantification for results on FB-7-21 in Table 2. Karger et al. (2024) do not publish individual LLM forecasts, precluding a direct pairwise comparison.

Forecaster	Brier score	Win Rate vs Best	Pairwise p-value
Public Survey (median per-event)	0.1510	0.2209	<0.001
Superforecasters (median per-event)	0.1152	0.5066	0.0946
Karger et al. (2024) state of the art	0.145	-	-
OpenAI o3	0.1262	0.4618	<0.001
AIA Forecaster	**0.1099**	-	-

Table 14: Uncertainty quantification for results on FB-8-14 in Table 2. Karger et al. (2024) do not publish individual LLM forecasts, precluding a direct pairwise comparison.

Forecaster	Brier score	Win Rate vs Best	Pairwise p-value
OpenAI o3	0.1324	0.7202	0.0016
AIA Forecaster	0.1258	0.7664	0.0567
Market	0.1106	-	-

Table 15: Uncertainty quantification for results on MarketLiquid in Table 2.

B. Evaluating Different Base Models

In this section we record the performance of different base models on the MarketLiquid. We choose this benchmark because it includes very recent events, all of which can be evaluated after the knowledge cutoff of the models we consider.[18] We exclude the supervisor agent (see Section 7.2) to isolate the quality of the baseline reasoning, and otherwise use the pipeline described in Section 3.

Model	Brier score	Win Rate vs Best	Pairwise p-value
OpenAI GPT-4o	0.1485	0.4416	<0.001
Gemini 2.5 Pro	0.1456	0.4099	<0.001
OpenAI GPT-5	0.1412	0.5115	<0.001
Qwen-32b	0.1404	0.3155	<0.001
OpenAI o3	0.1242	0.4422	0.1561
Claude Sonnet 4	0.1195	-	-

Table 16: Performance of various base models on MarketLiquid

As Table 16 shows, Sonnet 4 and OpenAI o3 yield comparable performance on the MarketLiquid; all other models perform significantly worse.[19] In addition to those presented above, we also find that OpenAI OSS-120b achieves a Brier score of 0.1076, with a win rate of 50% against Sonnet 4. However, closer inspection reveals that this

18. Claude Sonnet 4 has the latest knowledge cutoff, which is March 2025. However, its "reliable" knowledge cutoff is January 2025. The earliest questions in the MarketLiquid benchmark are asked "as of" 01/01/2025. This complicates the interpretation of the Sonnet 4 results, which likely include some form of lookahead bias (Sarkar and Vafa, 2024).
19. We find that Claude Opus 4 is prohibitively slow on our benchmarks and thus exclude it from our analysis.

apparently state-of-the-art performance is a mirage — the model is simply able to exploit the foreknowledge bias described in Section 5.2 more effectively than all other models we evaluate. Indeed, the analysis described in Section 5.2 reveals that 17.09% of the news queries leak foreknowledge about the event in question. These problematic queries cumulatively affect 57% of our benchmark, making it virtually impossible to evaluate the results.

We default to OpenAI o3 as our base model unless otherwise specified. Unlike Sonnet 4, its pretraining cutoff (October 2023) is well before the events which appear in Forecast-Bench and MarketLiquid, which precludes any form of pretraining lookahead bias on either benchmark (Sarkar and Vafa, 2024).

C. Sources of Foreknowledge Bias

Through extensive analysis of search API responses across Search-A and Search-B, we find that the primary mechanism for introducing foreknowledge bias is responses which return web pages or news articles that are updated after initial publication. They can be updated because of (1) live data feeds embedded on the page and/or (2) web pages that are continuously updated by the publisher.

Live data feeds that bypass API date filters. When search APIs crawl and summarize web pages, they often capture real-time widgets that display current data regardless of our specified date constraints. Consider this concrete example: we called the Search-A API requesting information about Tapestry Inc. (TPR) stock with an explicit July 21, 2024 cutoff date. Despite this restriction, the API returned:

> "As of July 23, 2025, TPR's price was approximately $108.72, meaning on July 21, 2024, the price was likely lower but near analyst forecasts of around $91.82..."

The API pulled this future price from a live widget on TipRanks, defeating the purpose of our temporal restriction. This occurs because the search APIs retrieve the current version of web pages, which often embed dynamically updating content.

Search APIs return the current versions of continuously updated pages. Wikipedia and similar platforms pose a distinct challenge: search APIs typically return the latest version of these pages, not a historical snapshot, even when a past date is specified.

Search APIs returning current versions of continuously updated pages. Wikipedia and similar platforms present a different challenge: when search APIs retrieve these pages, they get the most recent version rather than a historical snapshot, even when we specify a past date.